# Understanding Host Network Stack Overheads



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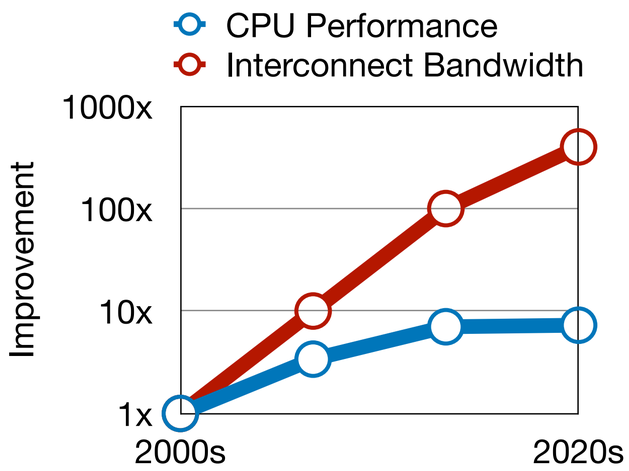

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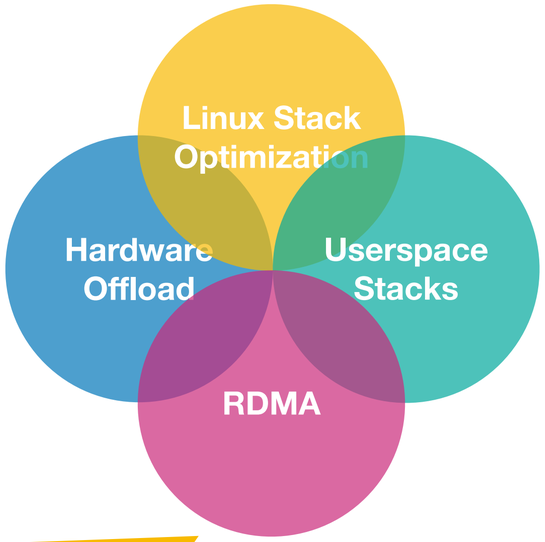
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# Network & End-host Hardware Trends




Exponentially Increasing Bandwidths.

Slowdown of Moore's Law.



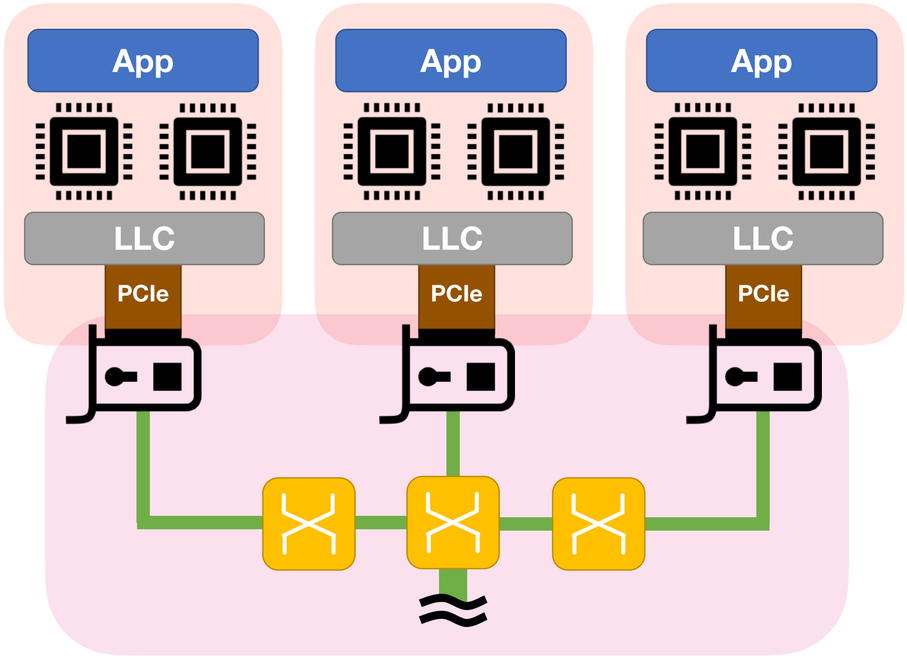

**| While hardware can support high-bandwidths, software has become the bottleneck.**

**| Our Goal: Guide the design-space by a detailed understanding of today's stacks.**

# Understanding End-Host Overheads in High-Bandwidth Links

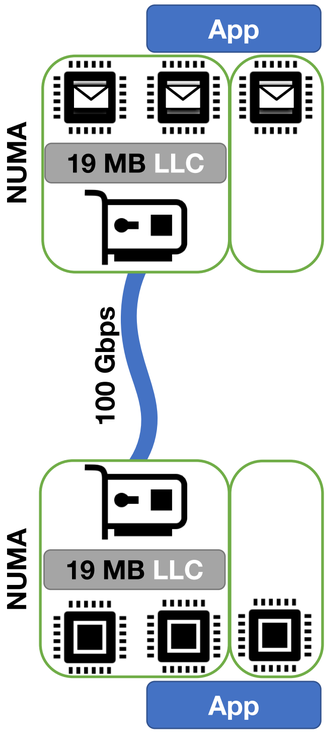
- In times of yore, for internet and the early generation datacenters, bottlenecks existed in the core-network.
  - Queuing Delay
  - Packet Drops
  - Switch Contention

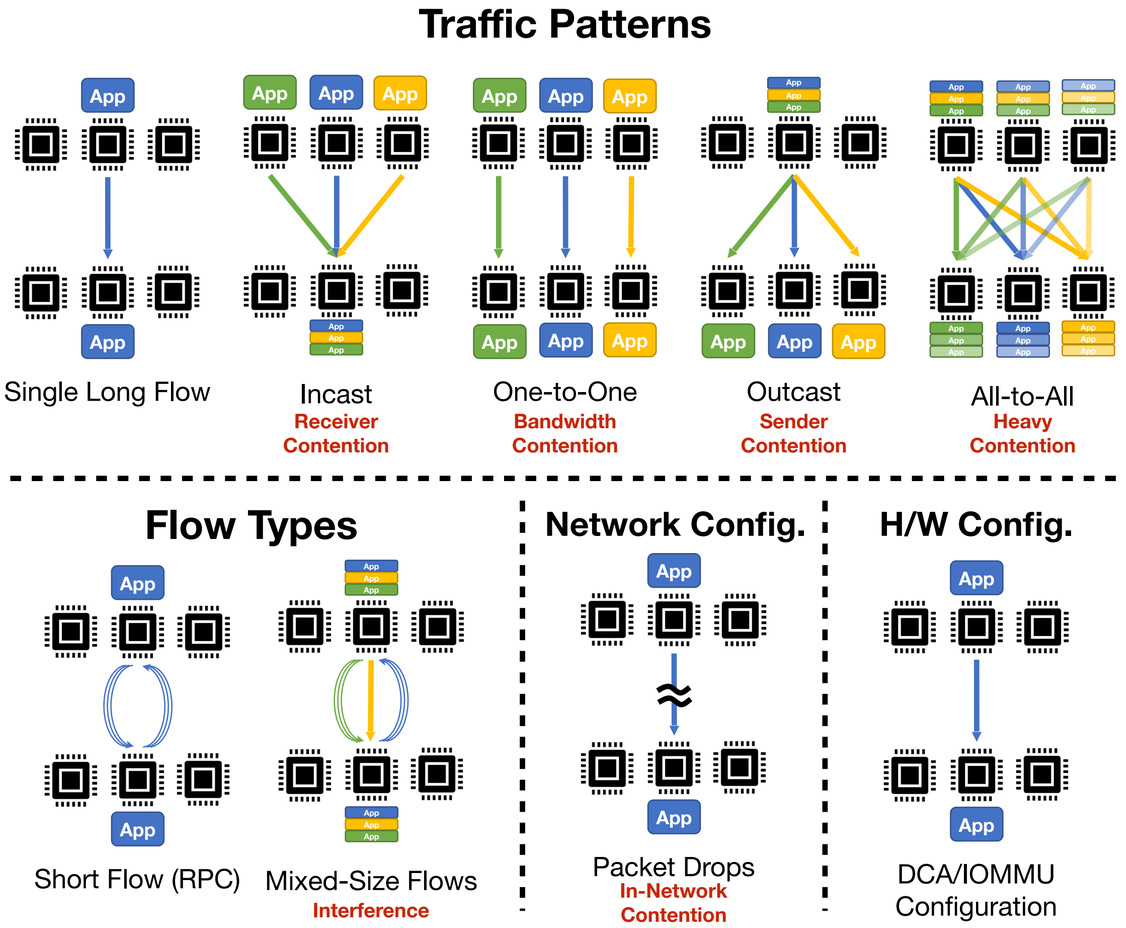
- Now, bottlenecks exist in both the core-network and end-hosts.
  - CPU Cores
  - DRAM Bandwidth
  - LLC Capacity



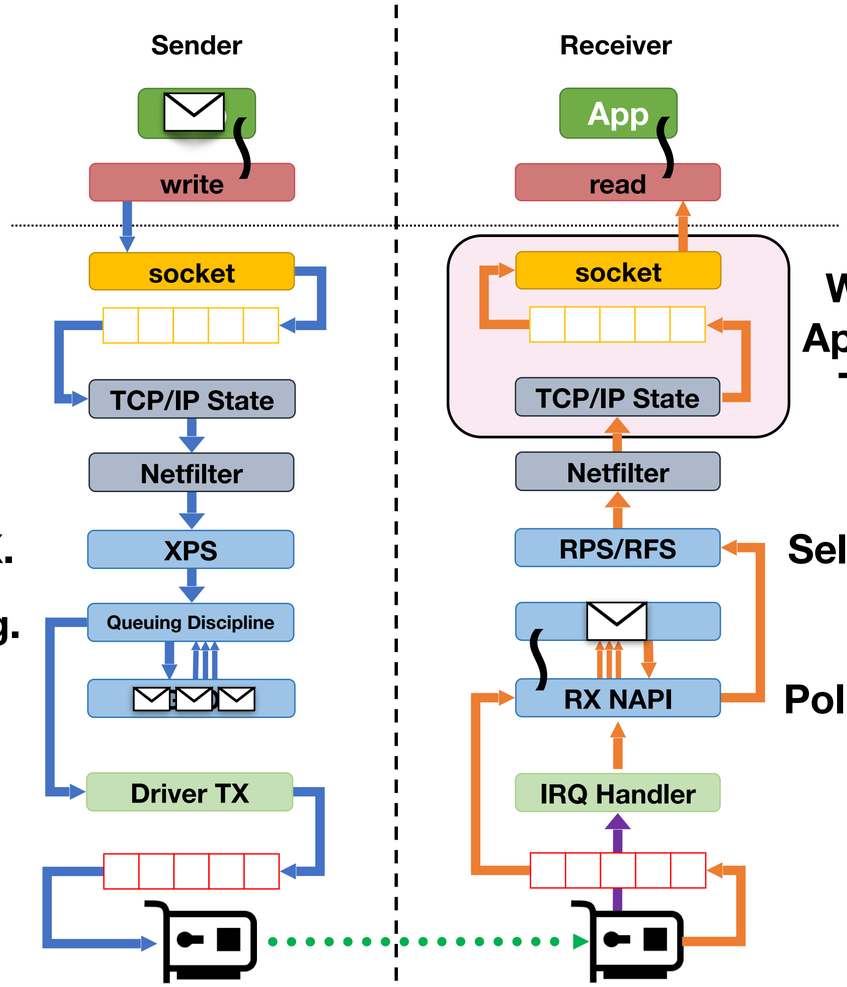
# Methodology & Experimental Scenarios

- DDIO can use ~20% LLC.
- Directly connected so bottleneck at the end-host.
- Run iPerf, Netperf to focus on the Linux network stack.
- Measure:
  - Throughput
  - CPU Utilisation & Breakdown
  - LLC Cache Miss Rate



# Linux Network Stack Data Path




Wake-up Application Thread.

Select the Hardware Queue for TX.

Packet Scheduling.

Select the CPU for TCP/IP Processing.

Poll for RX Packets.

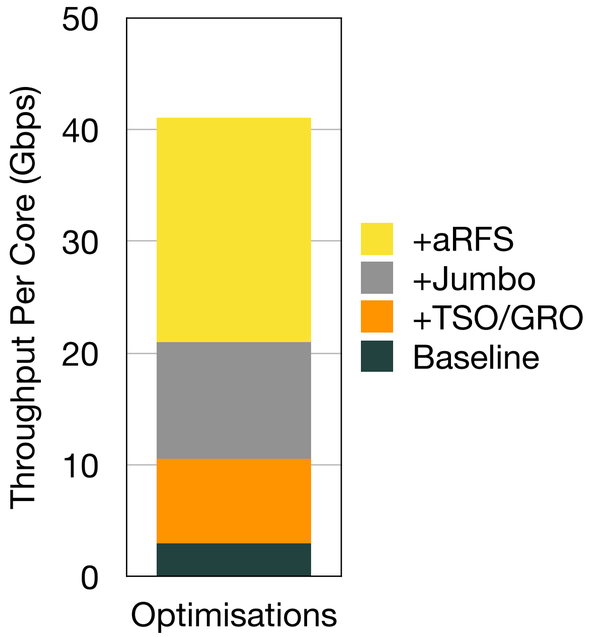
# Main Lessons from Our Study

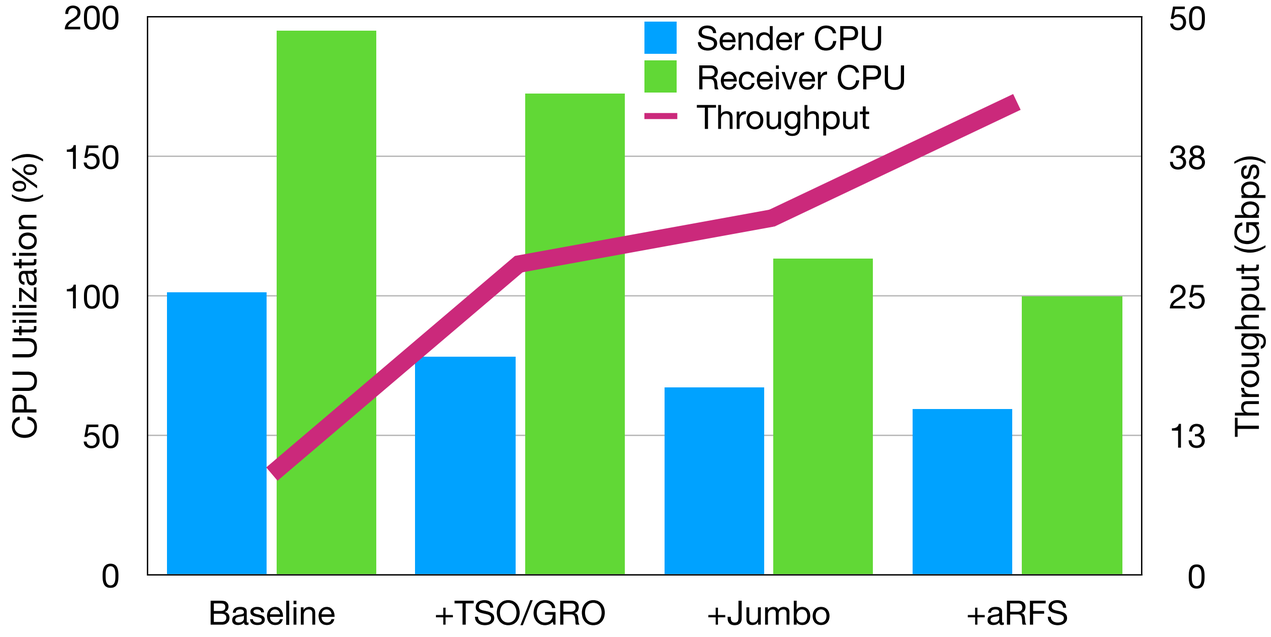
- Bottlenecks have shifted from packet processing to data copy.
  - More than 50% of all CPU cycles used for data copy in a single flow.
- The NIC DMA pipeline is inefficient.
  - As much as 24% drop in throughput when NIC DMA pipeline is suboptimal.
- End-host resource sharing considered harmful.
  - Flows sharing resources on a single CPU core causes up to 66% drop in throughput per core.
- Need redesigned network stacks and network aware CPU schedulers.
  - Long flows experience 52%, and short flows 57%, decrease in throughput per core when co-located.

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# Understanding Single Flow Performance



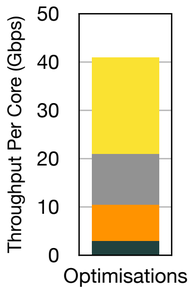



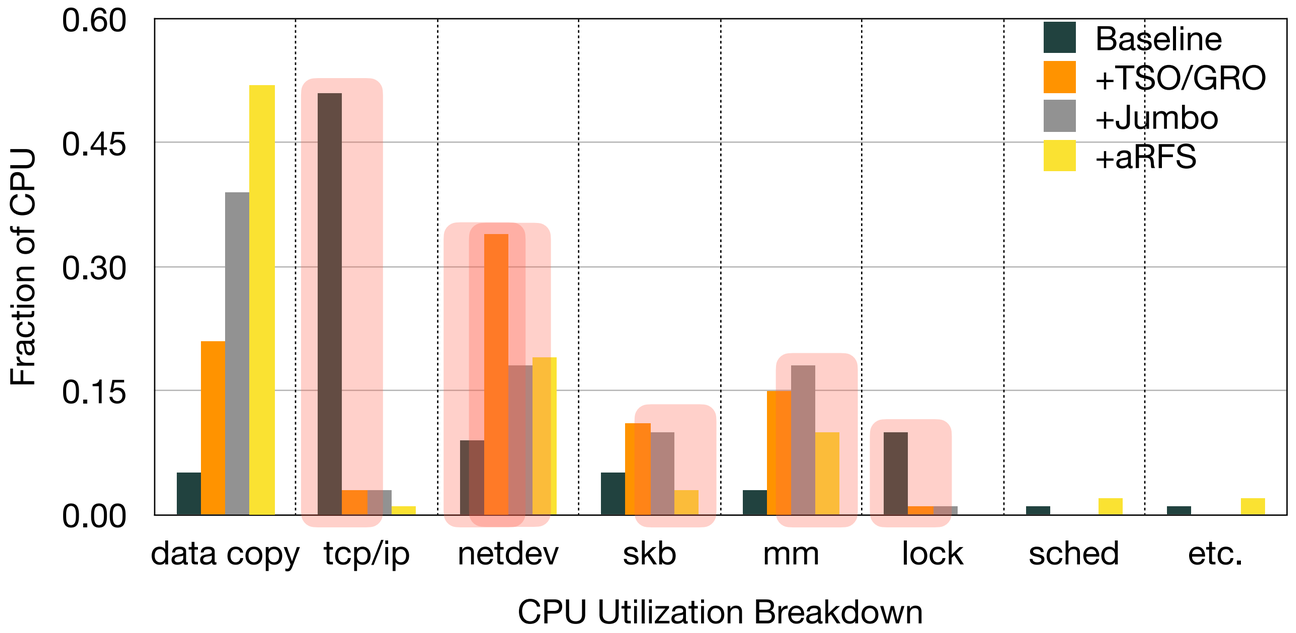

**Until 40 Gbps, a single CPU core could saturate the network bandwidth, is it still true for 100 Gbps?**

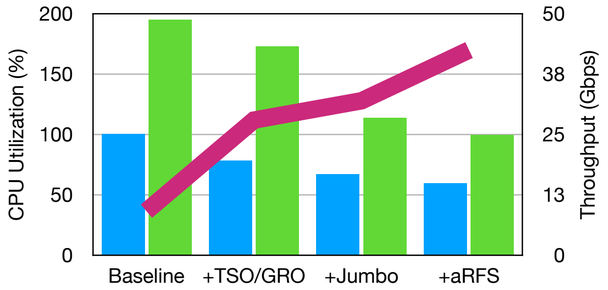
**Multiple cores needed to saturate NIC bandwidth.**

**Receiver side is the main bottleneck!**

# Deep Dive: Data Copy is the Main Bottleneck



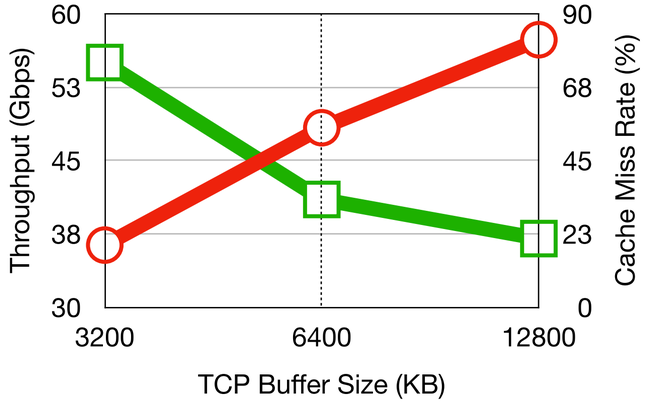



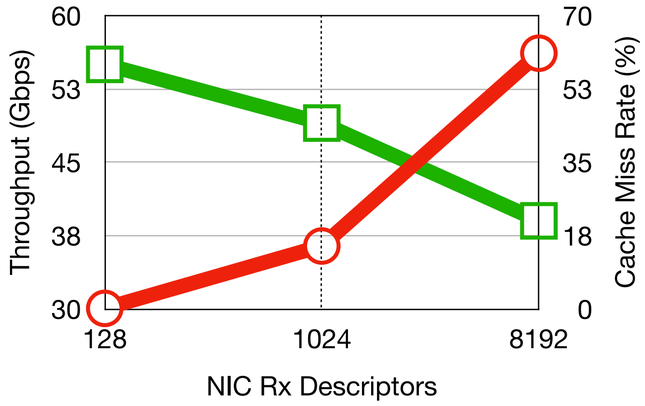

- GRO reduces overhead for TCP/IP and locking, NetDevice consumes more cycles.
- Jumbo frames reduce cost of GRO, overheads shift to data copy.
- aRFS improves cache locality, reducing cost of memory and `skb` management.

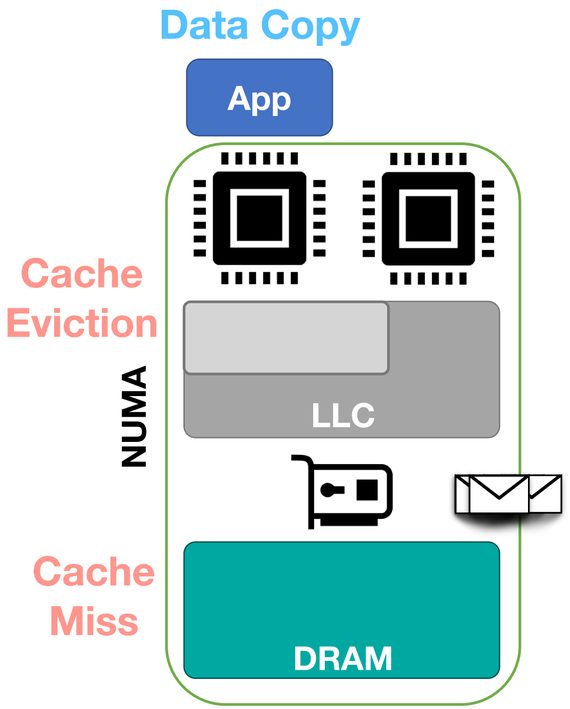
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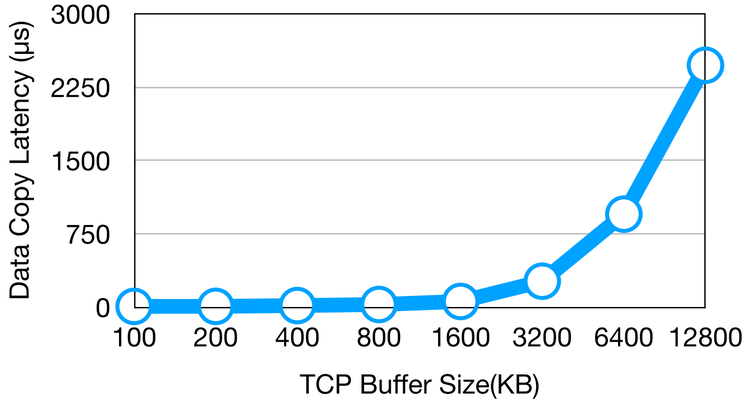
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# Deep Dive: Data Copy Overhead Due to Cache Misses



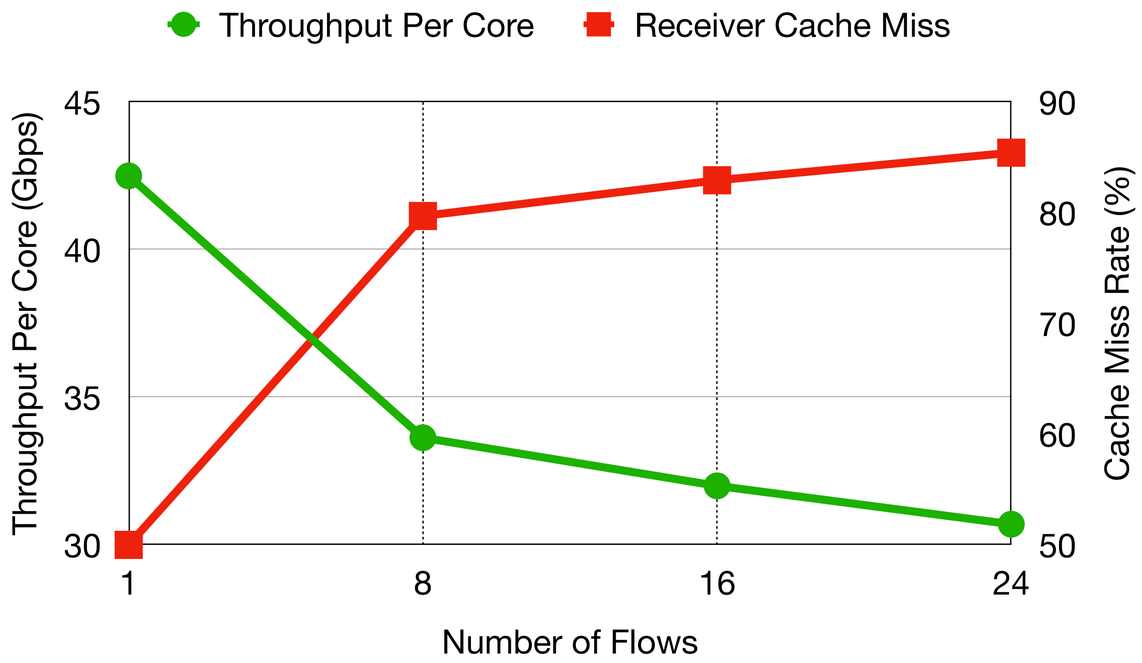

- Large TCP buffers increase data copy delay, which results in high cache miss rate.
- TCP buffer size calculation must take LLC size into account.
- More NIC Rx descriptors lead to a higher chance of cache eviction.
- We need a deeper understanding of this phenomenon.

- Zero-copy techniques like TCP mmap/AF_XDP alleviate data-copy overhead.
- Rearchitect network stack to scale different layers independently.

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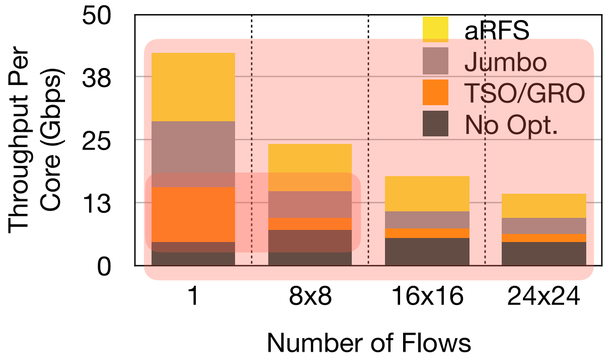
# Cache Contention vie Incast

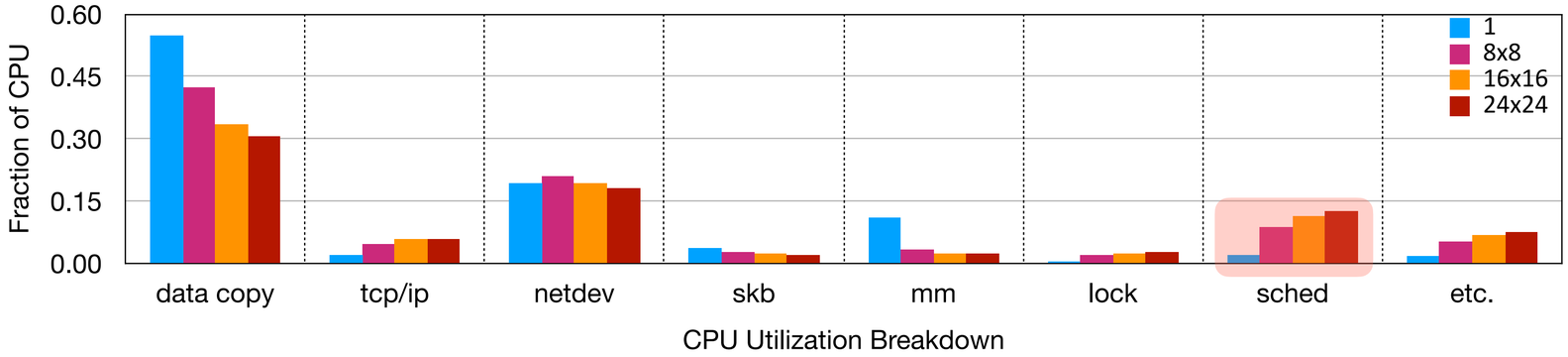


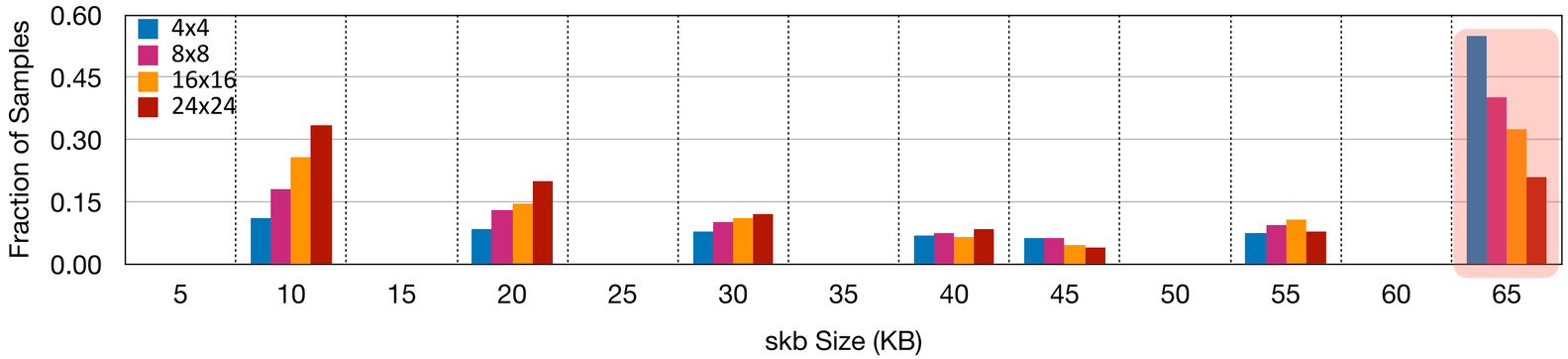

**When multiple flows contend for the CPU caches, throughput per core degrades.**

**Receiver driven transport protocols can be extended to orchestrate end-host resources like CPU caches.**

# Heavy Contention via All-to-All



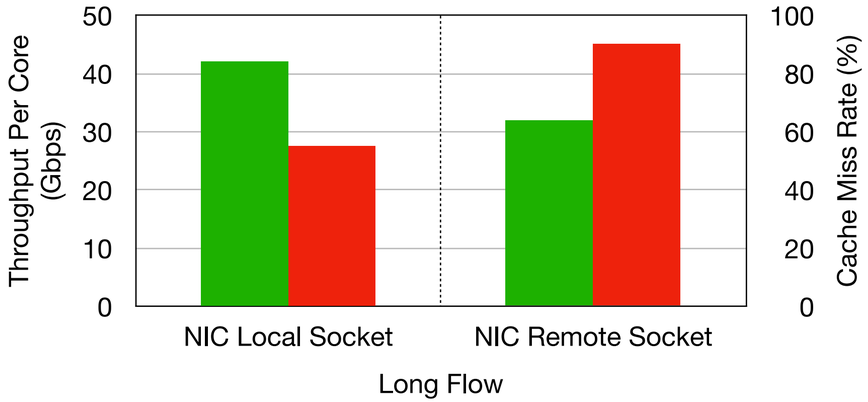



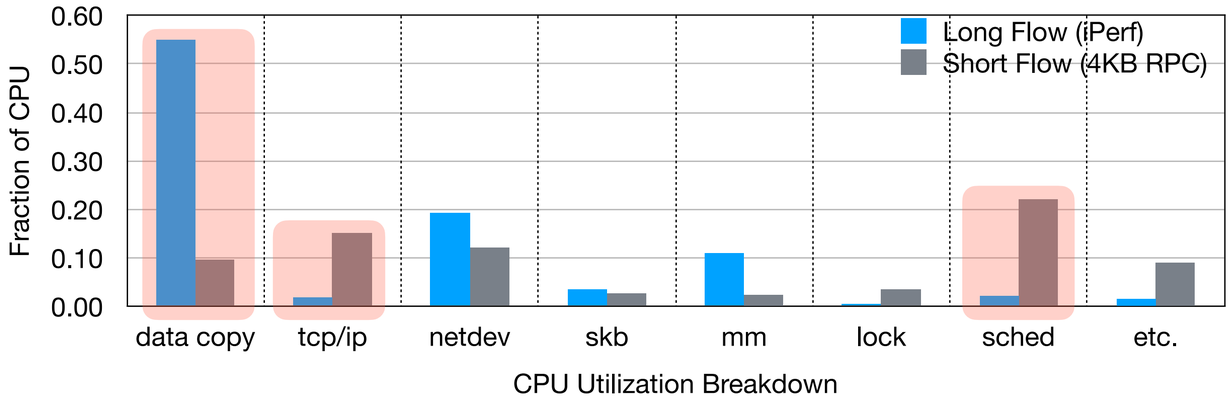

- **Throughput per core degrades by 67% due to contention for CPU and network.**
- **Scheduling overhead increases since applications sleep/wake-up frequently.**
- **GRO efficiency decreases due to many flows, increasing per packet overheads.**

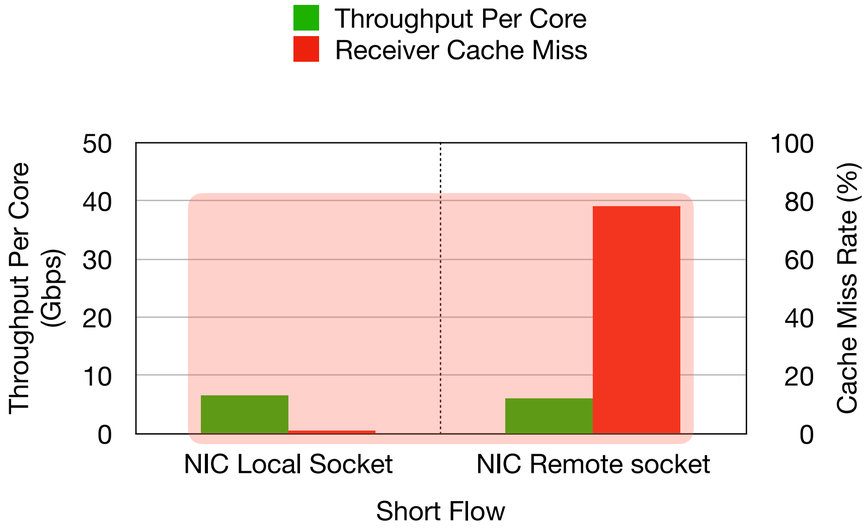
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# Difference in Bottlenecks for Long and Short Flows



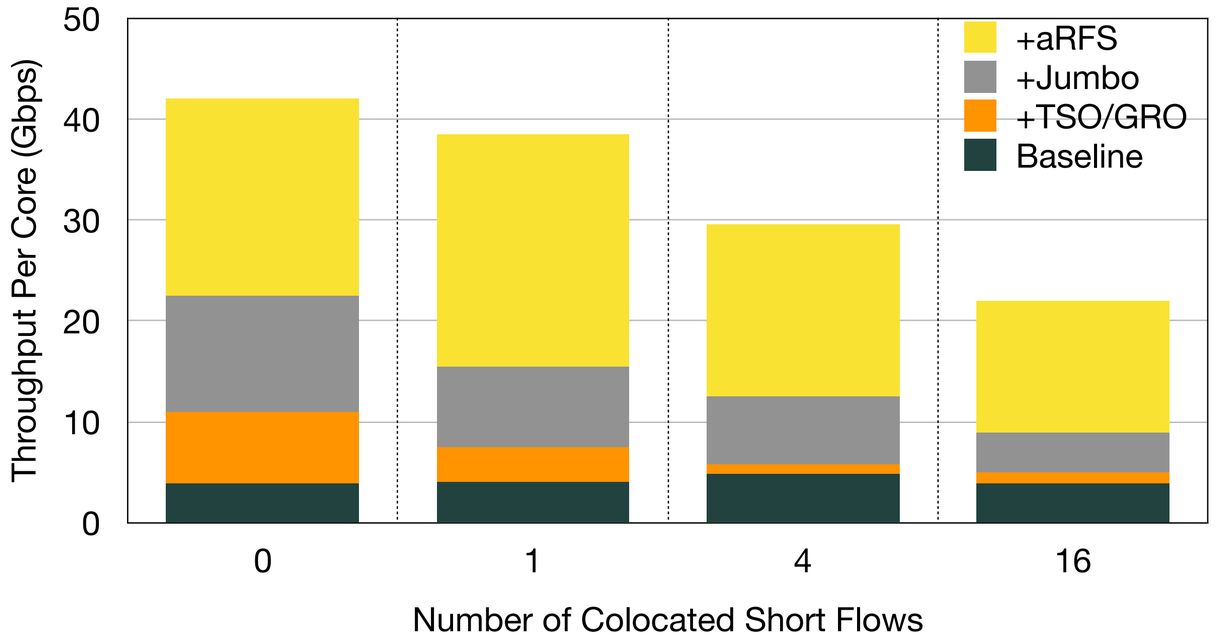




- **TCP/IP and scheduling overheads increase when flow size reduces.**
- **Design dedicated packet processing pipelines for short flows.**

- **DCA has little impact on the throughput of short flows.**
- **Schedule short flows on NIC remote NUMA.**

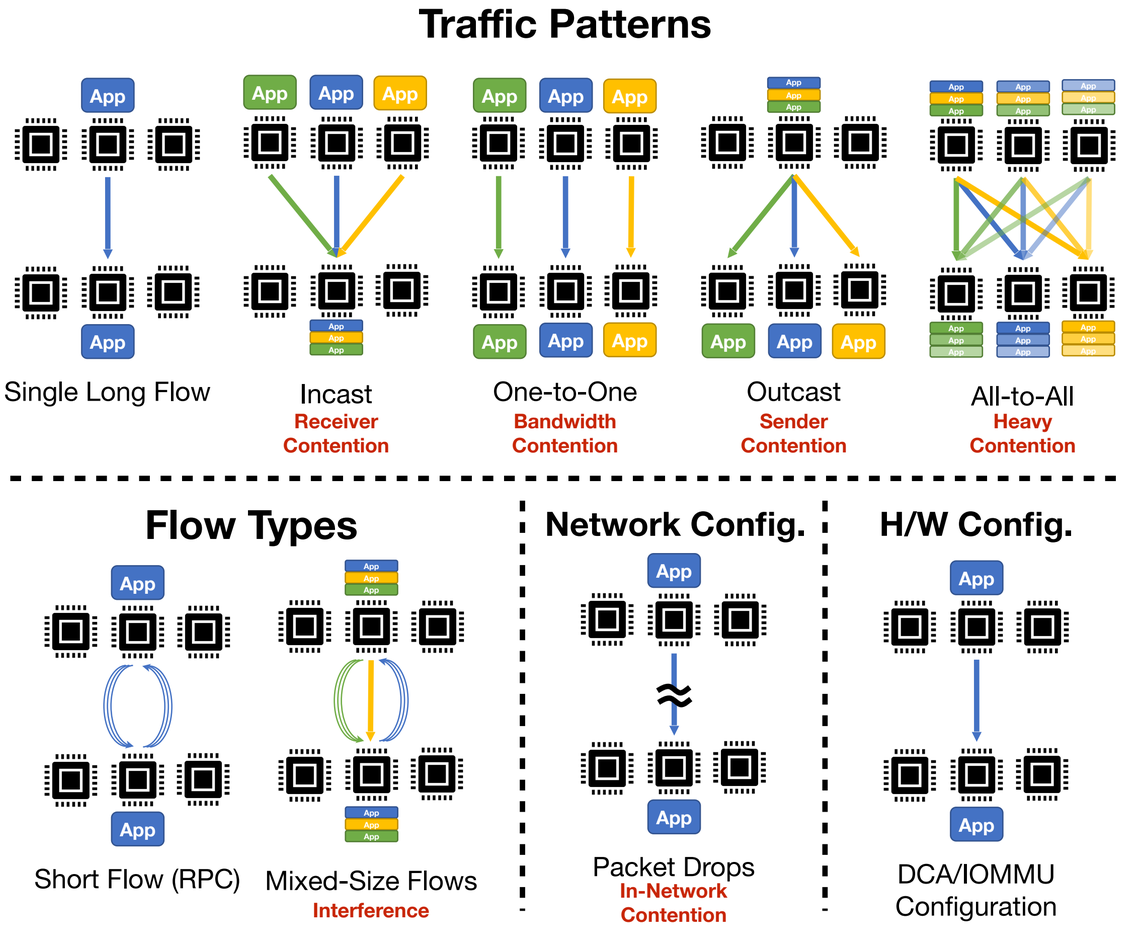
# Mixing Flow Sizes Considered Harmful




- **Overall throughput per core drops by 47%, long flow performance degrades by 52% and short flows 57%.**
- **Need network-aware CPU schedulers to schedule long and short flows on different cores and/or NUMA nodes.**

# More Results

- Please read the full paper for more details!
- The tools for profiling can be found at [https://github.com/Terabit-Ethernet/terabit-network-stack-profiling](https://github.com/Terabit-Ethernet/terabit-network-stack-profiling).



# Questions?